Decent Coaching Classes

VIII SB

Mind Your Math - IX

Time: 2 Hrs **Marks: 60**

Note: (1) All questions are compulsory. (2) Figures to the right indicate full marks. (3) Draw neat and proper figures wherever necessary.

Q.1 (A) Choose the correct alternative for each of the following questions. **[8]**

(1) Which of the following is an irrational number?

(A) $\sqrt{49}$ (B) 22/7 (C) $\sqrt{3}$ (D) 0.75

(2) Which of the following comparisons is correct?

(A) $-2/3 > -3/4$ (B) $-2/3 < -3/4$ (C) $-2/3 = -3/4$ (D) $-2/3 < -5/4$

(3) Two parallel lines are intersected by a transversal. If one of the interior angles on the same side of the transversal is 75°, the other interior angle is ___

(A) 75° (B) 105° (C) 15° (D) 115°

(4) Line l ∥ line m and line t is their transversal. A pair of corresponding angles measures $(2x + 10)°$ and $(3x - 20)°$. The value of x is ___

(A) 20 (B) 25 (C) 30 (D) 35

(5) The value of $\sqrt[3]{0.000343}$ is ___

(A) 0.7 (B) 0.07 (C) 0.007 (D) 7

(6) The value of $32^{3/5}$ is ___

(A) 6 (B) 8 (C) 16 (D) 2

(7) The point of concurrence of the three altitudes of a triangle is called the ___

(A) centroid (B) incentre (C) circumcentre (D) orthocentre

(8) In ΔABC, seg AD is a median and G is the centroid. If l(AD) = 18 cm, then l(AG) = ___

(A) 6 cm (B) 9 cm (C) 12 cm (D) 15 cm

Q.1 (B) Fill in the blanks (1 to 5) and state whether the following are true or false (6 to 8). **[8]**

(1) The decimal form of an irrational number is non-terminating and _____.

(2) A line which intersects two given lines in two distinct points is called a _____ of those lines.

(3) The fifth root of 243 written in index form is _____.

(4) The orthocentre of a right-angled triangle is the vertex of the _____ angle.

(5) The orthocentre of an obtuse-angled triangle lies in the _____ of the triangle.

(6) $3 + \sqrt{2}$ is an irrational number. (True / False)

(7) The cube root of −27 is 3. (True / False)

(8) If x varies inversely as y, then $x \div y$ is a constant. (True / False)

Q.2 Solve any FIVE of the following sub-questions. **[10]**

(1) Expand: $(2p + 5)(2p - 3)$

(2) Expand: $(y + 5)^3$

(3) Factorise: $x^2 + 11x + 28$

(4) Factorise: $x^3 - 125$

(5) m varies directly as n. When m = 12, n = 4. Find the constant of variation and write the equation of variation.

(6) The diagonals of a rhombus are 24 cm and 10 cm. Find the length of its side.

(7) The marked price of a school bag is ₹ 1500. A discount of 12% is given on it. Find the selling price of the bag.

Q.3 Solve any FOUR of the following sub-questions. [12]

(1) Expand: $(3x - 2y)^3$

(2) Expand: $(2p + q + 3)^2$

(3) Factorise: $2x^2 + 7x - 15$

(4) Factorise: $54a^3 - 2$

(5) 12 workers can finish a piece of work in 20 days. How many workers are needed to finish the same work in 15 days?

(6) A farmer sold 25 quintals of wheat at ₹ 2400 per quintal through an agent. The agent charged 3% commission. Find the commission paid and the amount the farmer received.

Q.4 Solve any THREE of the following sub-questions. [12]

(1) Simplify: $(m + 3)^3 - (m - 3)^3$

(2) Simplify: $(x^2 - 5x + 6) / (x^2 - 9)$

(3) p varies inversely as the square of q. When q = 4, p = 5. Find (i) p when q = 2, (ii) q when p = 1.25.

(4) Construct □ABCD such that l(AB) = 5 cm, l(BC) = 4.5 cm, m∠A = 70°, m∠B = 100°, m∠C = 110°.

(5) A shopkeeper marks the price of an article 30% above the price he decided. While selling it he gives a discount of 10%. Find the percentage gain on the decided price.

Q.5 Solve any TWO of the following sub-questions. [10]

(1) In parallelogram PQRS, the measures of adjacent angles ∠P and ∠Q are $(3x + 10)°$ and $(2x + 20)°$ respectively. Find (i) the value of x, (ii) the measures of all four angles of the parallelogram, (iii) l(QT), if the diagonals meet at T and l(QS) = 14 cm.

(2) A shopkeeper gives 8% discount on an article whose marked price is ₹ 2500 and still earns a profit of 15%. Find the cost price of the article for the shopkeeper.

(3) (i) Using an expansion formula, find the value of $(102)^3$. (ii) Factorise: $6x^2 + x - 15$.

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